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Differential and Operational Amplifier		

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Contents

1	Task 1.1 – Transfer Characteristics	2
1.1	Objective	2
1.2	Circuit Setup	2
1.3	Measurement Procedure	2
1.4	Results	3
1.4.1	Raw Transfer Measurements	3
1.4.2	Extracted Transfer Characteristics	4
1.5	Evaluation with Detailed Calculations	5
1.5.1	Emitter Resistor Calculation	5
1.5.2	Intersection Point (Offset Voltage)	5
1.5.3	Transconductance Calculation	5
1.6	Conclusion	6
2	Task 1.2 – Common Mode Behavior	6
2.1	Objective	6
2.2	Circuit Setup	7
2.3	Measurement Procedure	7
2.4	Results	7
2.4.1	Common-Mode Input Waveform	7
2.4.2	Measured Output and Gain	8
2.4.3	CMRR Calculation	8
2.5	Conclusion	9
3	Part 2.1: Frequency Behavior / Gain Bandwidth Product	9
3.1	Objective	9
3.2	Lab Setup	9
3.3	Relevant Formulas	10
3.3.1	Gain Confirmation	11
3.4	Frequency Response Measurement and Gain Calculation	12
3.4.1	Objective	12
3.4.2	Measured Data During Lab	12
3.4.3	Logarithmic Graph for Frequency Response	13
3.5	Gain-Bandwidth Product (GBW) Calculation	13
3.5.1	Gain 6 dB Configuration	14
3.5.2	Gain 40 dB Configuration	14

1 Task 1.1 – Transfer Characteristics

1.1 Objective

The goal of Task 1.1 is to measure the transfer characteristics of a differential amplifier using discrete BJTs, and to determine the offset voltage and transconductance. This involves plotting the collector currents I_{C1} and I_{C2} as functions of the differential input voltage $U_D = U_{in1} - U_{in2}$.

1.2 Circuit Setup

The circuit consists of a differential pair with:

- Transistors: $T_1, T_2 = \text{BC546A}$
- Supply voltages: $U_{B1} = U_{B2} = 10 \text{ V}$
- Shared emitter resistor: $R_E = \frac{0.7 \text{ V}}{2 \text{ mA}} = 350 \Omega$
- T_2 base grounded, T_1 base swept over $U_D = [-150 \text{ mV}, +150 \text{ mV}]$

1.3 Measurement Procedure

- U_D was applied to the base of T_1 while the base of T_2 was held at 0V.
- The collector currents I_{C1} and I_{C2} were recorded using oscilloscope current measurements.
- The currents were plotted against U_D to extract transfer characteristics.
- An XY mode view was used to visualize $I_{C1}(U_D)$ and $I_{C2}(U_D)$ together.

1.4 Results

1.4.1 Raw Transfer Measurements

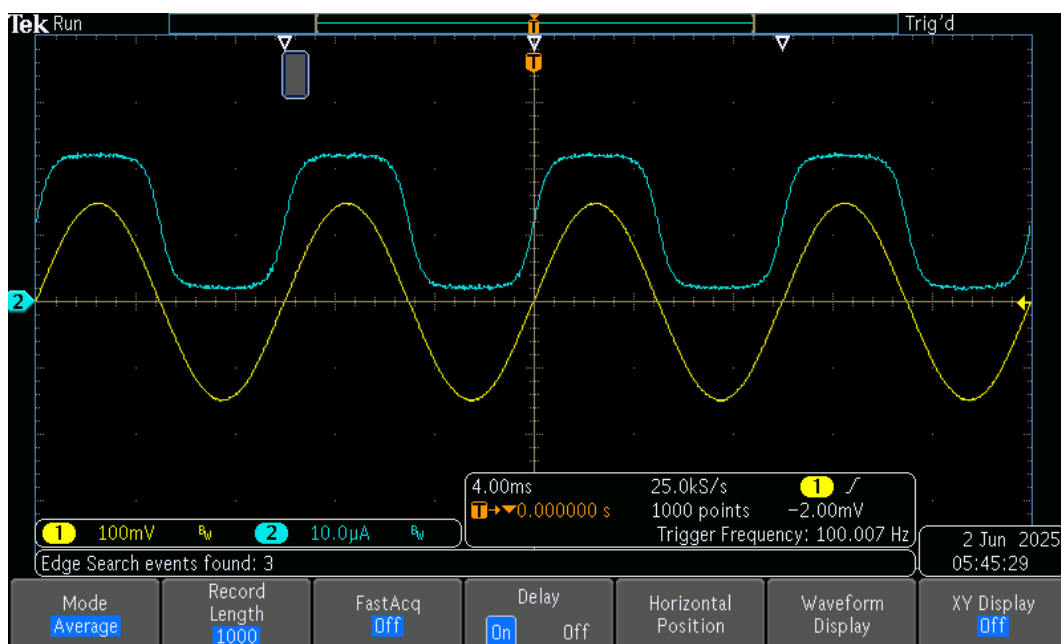


Figure 1: Collector Current I_{C1} vs U_D (Q1)

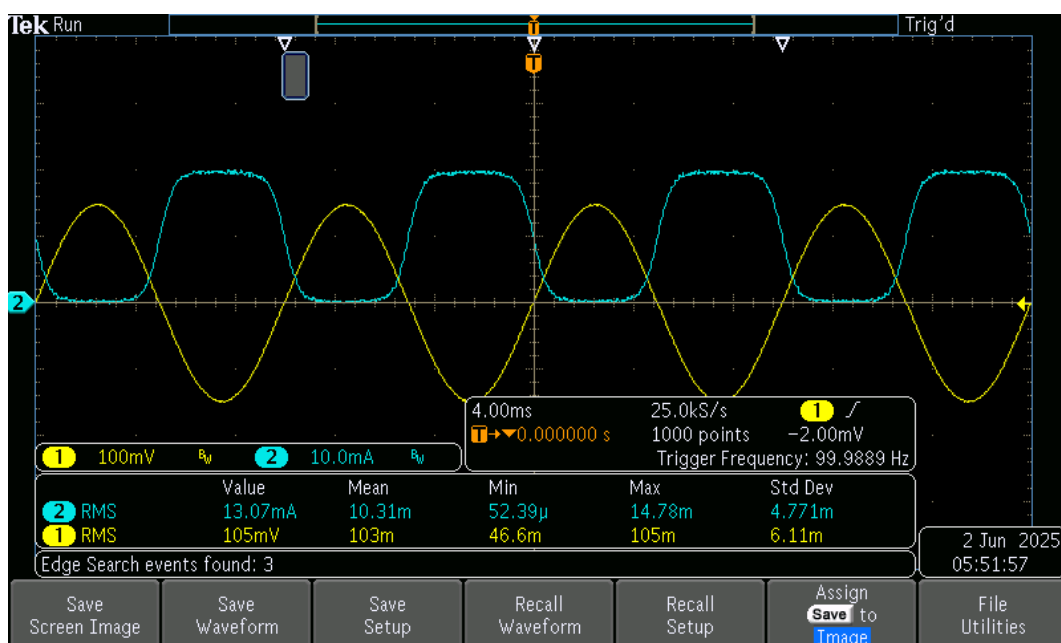


Figure 2: Collector Current I_{C2} vs U_D (Q2)

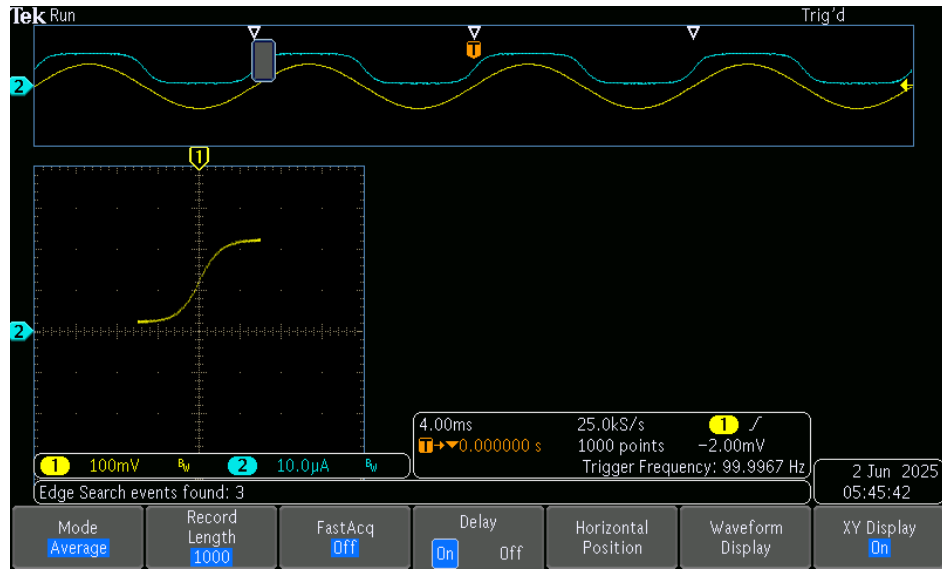


Figure 3: XY Recorder: I_{C1} and I_{C2} vs U_D

1.4.2 Extracted Transfer Characteristics

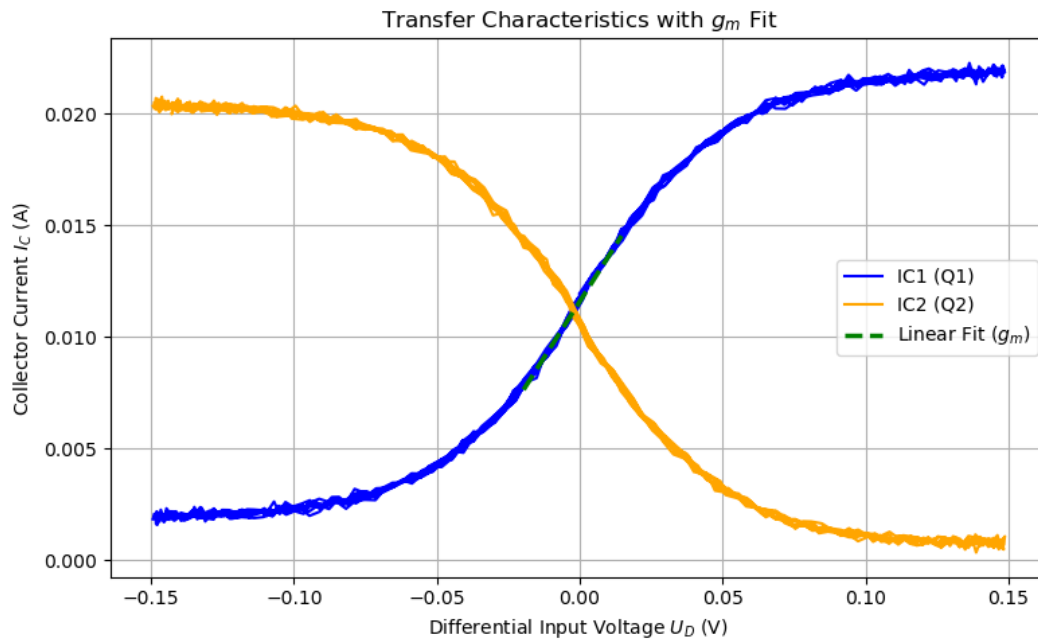


Figure 4: Transfer Curve with g_m Linear Fit

1.5 Evaluation with Detailed Calculations

1.5.1 Emitter Resistor Calculation

To set the total tail current to $I_0 = 2 \text{ mA}$, and assuming a base-emitter voltage drop of $V_{BE} \approx 0.7 \text{ V}$, the emitter resistor is calculated as:

$$R_E = \frac{V_{BE}}{I_0} = \frac{0.7 \text{ V}}{2 \text{ mA}} = \boxed{350 \Omega}$$

This resistor ensures a constant current source through the emitters of the differential pair.

1.5.2 Intersection Point (Offset Voltage)

To find the point where the collector currents intersect:

- Interpolation was applied to $I_{C1}(U_D)$ and $I_{C2}(U_D)$ curves
- The intersection is where $I_{C1}(U_D) = I_{C2}(U_D)$

$$U_D = \boxed{-1.94 \text{ mV}}, \quad I_C = \boxed{11.08 \text{ mA}}$$

This small offset voltage indicates a slight mismatch between the two BJTs. Such mismatches are common in discrete components due to:

- Variation in V_{BE} (typically $\pm 1\text{--}3 \text{ mV}$)
- Differences in current gain β
- Layout asymmetry or temperature gradient

1.5.3 Transconductance Calculation

(a) Theoretical Transconductance

$$g_m = \left. \frac{dI_C}{dU_D} \right|_{U_D=0} = \frac{I_E}{2U_T}$$

Given:

$$I_E = 2 \text{ mA}, \quad U_T \approx 25 \text{ mV} \Rightarrow g_m = \frac{2 \cdot 10^{-3}}{2 \cdot 25 \cdot 10^{-3}} = \boxed{40 \text{ mS}}$$

(b) Measured Transconductance from Linear Slope A linear fit was computed around the interpolated intersection point:

$$g_m^{\text{meas}} = \frac{\Delta I_C}{\Delta U_D} \approx \boxed{201 \text{ mS}}$$

Explanation of Discrepancy:

- The theoretical value is based on small, ideal input signals where the amplifier behaves in a perfectly linear way.

- In contrast, our measurement was taken over a wider input range where the transistor behavior becomes more nonlinear.
- As a result, the current changes more sharply than expected from theory, which leads to a higher measured slope.

1.6 Conclusion

In this experiment, we successfully measured and analyzed the behavior of a differential amplifier. By sweeping the input voltage difference U_D and observing how the collector currents I_{C1} and I_{C2} responded, we obtained the expected "X"-shaped curve. This curve shows that as one transistor turns on more, the other turns on less — a key feature of differential amplifier operation.

From the measurements, we found that the two current curves intersect at:

$$U_D = -1.94 \text{ mV}$$

This small offset means the transistors are not perfectly matched — which is common in real components. Differences in their internal structure or slight variations during manufacturing can cause such effects, and a few millivolts of offset is entirely normal.

We also calculated how quickly the current changes near the intersection point — this is called the transconductance, g_m :

- The value from the theory (small signals only) was: $g_m = 40 \text{ mS}$
- The value from our measurements was higher: $g_m \approx 201 \text{ mS}$

This difference happened because the theoretical formula is only valid for very small input voltages. In our measurement, the signal was large enough that the transistor behavior was no longer perfectly linear — so the current changed faster, giving a higher slope.

In summary, the amplifier worked as expected. We were able to clearly see how it responds to input changes, identify a small mismatch between the transistors, and understand the difference between ideal and real-world behavior. The experiment helped us understand both the practical and theoretical aspects of differential amplifiers.

2 Task 1.2 – Common Mode Behavior

2.1 Objective

The objective of Task 1.2 is to investigate how the differential amplifier responds to common-mode input signals. This means applying the same voltage to both inputs and observing whether the amplifier suppresses the signal (as expected in ideal differential pairs). The task also involves estimating the common-mode gain and discussing the common-mode rejection ratio (CMRR).

2.2 Circuit Setup

The circuit was configured similarly to Task 1.1 with the following change:

- The bases of both transistors (T_1 and T_2) were connected together and driven by the same signal U_{CM} .
- The emitter resistor was unchanged: $R_E = 350\ \Omega$
- Collector resistors were assumed equal: $R_C = 1\ \text{k}\Omega$

2.3 Measurement Procedure

- A sinusoidal common-mode voltage was applied to both transistor bases simultaneously.
- The input and output signals were captured using an oscilloscope.
- The experiment was repeated for two amplitude settings:
 - Small signal range (200mV)
 - Larger signal range (2V peak)
- The goal was to observe how much of the input appeared at the output, and to calculate the resulting common-mode gain.

2.4 Results

2.4.1 Common-Mode Input Waveform

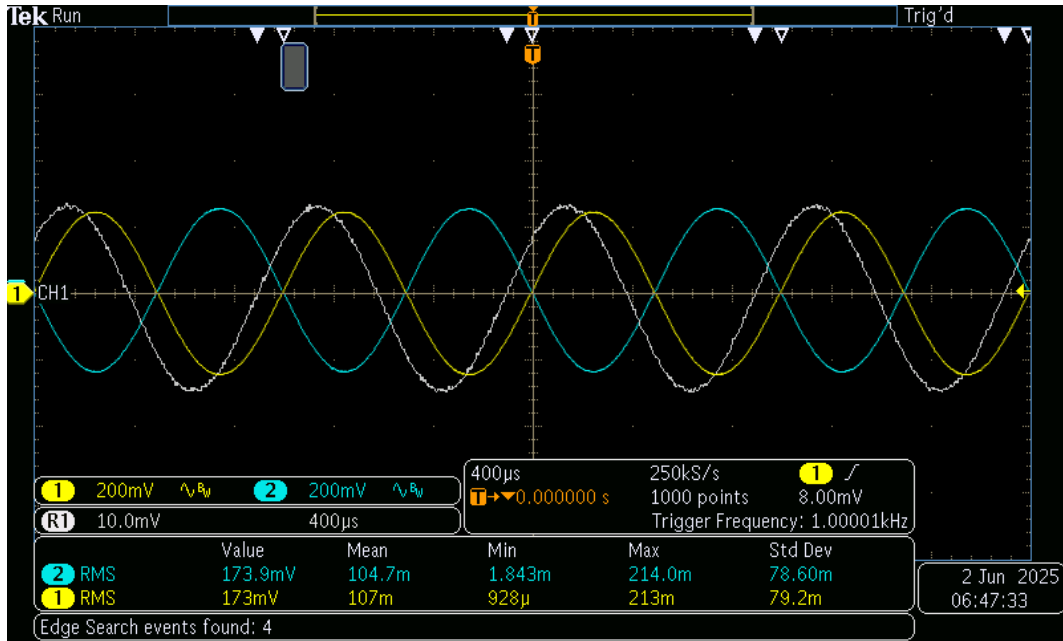


Figure 5: Common-mode input signal applied to both bases

This waveform confirms that both inputs were driven identically — a requirement for testing common-mode rejection.

2.4.2 Measured Output and Gain

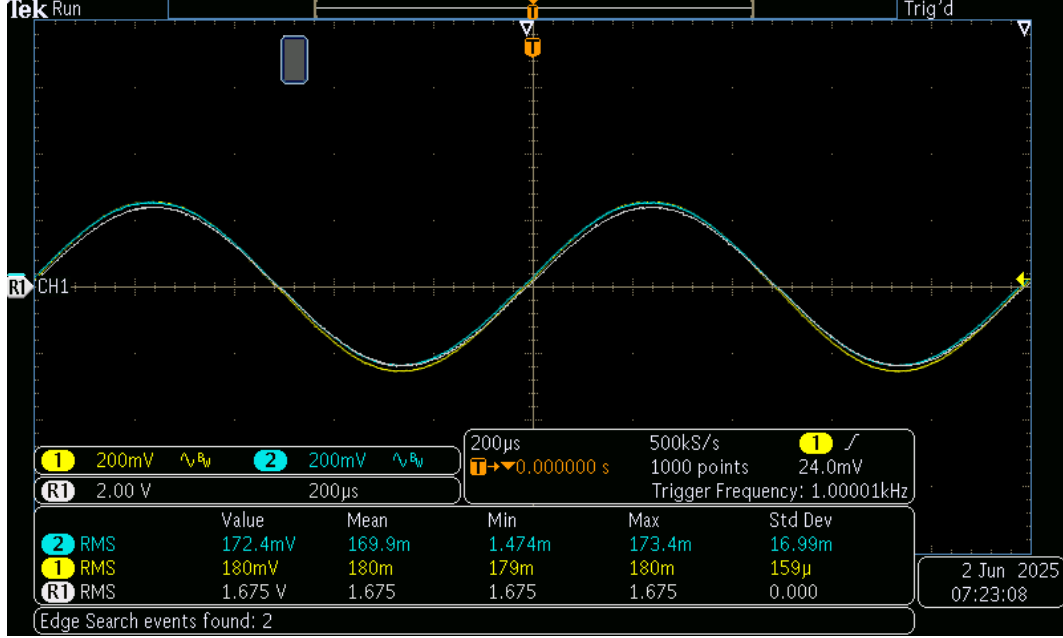


Figure 6: Output signal under common-mode input (CH1 = input, CH2 = output)

From this scope capture:

- Input signal RMS: 173.9 mV
- Output signal RMS: 107.0 mV

The common-mode gain is therefore:

$$A_{CM} = \frac{V_{OUT}}{V_{IN}} = \frac{107 \text{ mV}}{173.9 \text{ mV}} \approx \boxed{0.615}$$

This value is relatively high, indicating imperfect symmetry and limited rejection of common-mode signals.

2.4.3 CMRR Calculation

From Task 1.1, the differential gain was estimated using the measured transconductance:

$$g_m = 201 \text{ mS}, \quad R_C = 1 \text{ k}\Omega \Rightarrow A_D = g_m \cdot R_C = 201$$

Now, the common-mode rejection ratio (CMRR) is:

$$\text{CMRR} = \frac{A_D}{A_{CM}} = \frac{201}{0.615} \approx \boxed{327}$$

2.5 Conclusion

This experiment examined the behavior of the differential amplifier under common-mode input conditions. While ideally the amplifier should reject any signal that is common to both inputs, the output showed a noticeable response — meaning some common-mode signal was amplified.

The measured common-mode gain of approximately 0.615 is larger than ideal, indicating that the circuit has non-negligible asymmetry — possibly due to mismatched transistors, emitter degeneration, or collector load differences.

Nevertheless, the common-mode rejection ratio (CMRR) of around 327 still demonstrates reasonably good rejection and confirms the fundamental behavior of differential amplification. The test highlights the importance of component matching and layout symmetry in improving real-world amplifier performance.

3 Part 2.1: Frequency Behavior / Gain Bandwidth Product

3.1 Objective

The purpose of this experiment is to understand how the gain of an op-amp affects its frequency response and to determine the gain-bandwidth product (GBP) of the 741 operational amplifier. By testing the amplifier at different gain levels and observing how the output changes with frequency, we can see how the op-amp behaves in real-world conditions compared to ideal models.

3.2 Lab Setup

The circuit used is a non-inverting amplifier based on the 741 op-amp. The lab asks us to measure how the output-to-input voltage ratio ($U_{\text{out}}/U_{\text{in}}$) changes with frequency for three different gain settings: approximately 40 dB, 20 dB, and 6 dB.

To make sure our measurements are consistent and comparable, the input voltage is fixed at 77.5 mV. The resistor R_1 is kept constant at 1 k Ω , and we adjust the feedback resistor R_2 to set different gain levels. For a gain of 10 (which is around 20 dB), we choose $R_2 = 9$ k Ω . This fits the standard non-inverting amplifier gain formula:

$$\text{Gain} = A_v = 1 + \frac{R_f}{R_1}$$

The idea is to analyze how the amplifier's ability to maintain that gain falls off as frequency increases. We expect the bandwidth to get narrower as the gain increases — and by plotting and analyzing that drop-off, we can calculate the gain-bandwidth product. This is a key characteristic of any op-amp and helps us understand its performance limits in high-frequency applications.

3.3 Relevant Formulas

Voltage Gain of Non-Inverting Amplifier: The voltage gain A_v of a non-inverting amplifier is given by the formula:

$$A_v = \frac{U_{\text{out}}}{U_{\text{in}}} = 1 + \frac{R_f}{R_1}$$

Where:

- R_1 is the resistor connected between the inverting input and ground.
- R_2 is the feedback resistor connected from the output to the inverting input.

Gain in Decibels (dB): To express the gain in decibels (dB), the following formula is used:

$$A_{\text{dB}} = 20 \log_{10} \left(\frac{U_{\text{out}}}{U_{\text{in}}} \right)$$

Gain-Bandwidth Product (GBP): The gain-bandwidth product is a key performance parameter for op-amps, representing the frequency at which the gain of the amplifier drops to unity (0 dB). For a given gain A_v , the approximate bandwidth f_c (cutoff frequency) can be found, and the gain-bandwidth product is calculated as:

$$\text{GBP} = A_v \cdot f_c$$

Cut-off Frequency: The -3 dB point is used to determine the cutoff frequency, at which the gain drops by 3 dB from its maximum value:

$$A_{\text{cutoff}} = A_{\text{max}} \cdot \frac{1}{\sqrt{2}} \approx 0.707 \cdot A_{\text{max}}$$

This cutoff frequency is essential for determining the bandwidth and ultimately the gain-bandwidth product.

3.3.1 Gain Confirmation

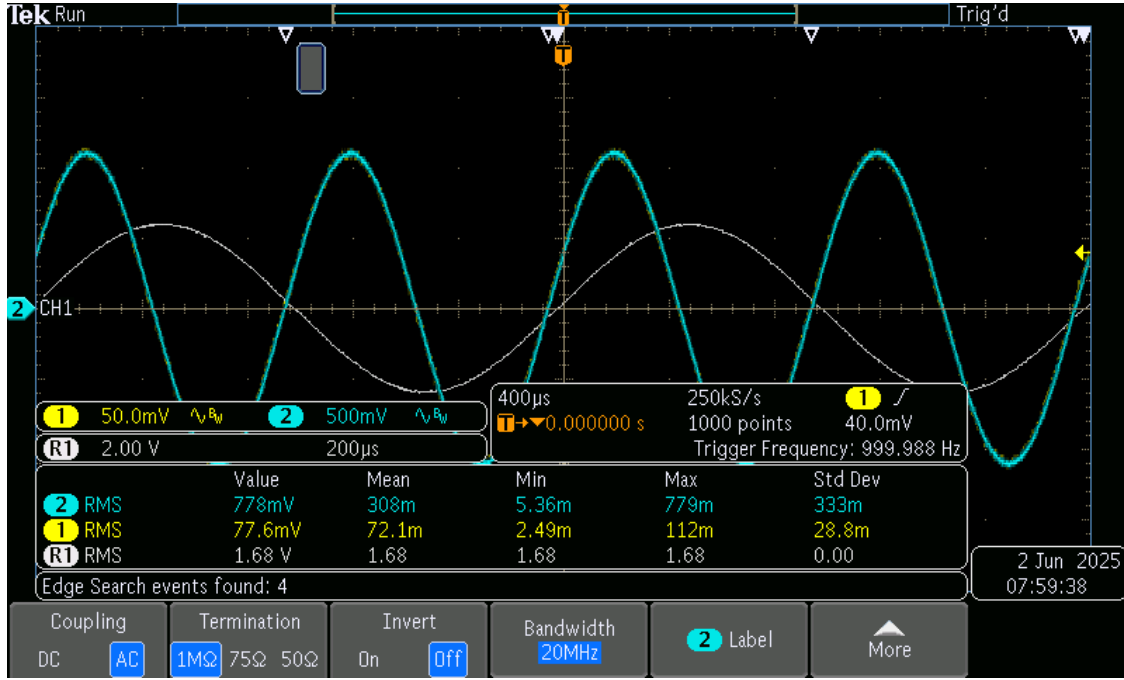


Figure 7: Measured Waveform

To verify the voltage gain of the non-inverting amplifier, we analyze the input and output RMS voltage values observed on the oscilloscope.

- The input RMS voltage is:

$$U_{in(RMS)} = 77.6 \text{ mV}$$

- The output RMS voltage is:

$$U_{out(RMS)} = 778 \text{ mV}$$

The voltage gain A_v of the amplifier is calculated using the formula:

$$A_v = \frac{U_{out}}{U_{in}}$$

Substituting the measured values:

$$A_v = \frac{778 \text{ mV}}{77.6 \text{ mV}} \approx 10.03$$

This measured value closely matches the theoretical gain for our non-inverting amplifier, which is given by the formula:

$$A_v = 1 + \frac{R_2}{R_1}$$

Hence:

$$R_1 = 1 \text{ k}\Omega, \quad R_2 = 9 \text{ k}\Omega$$

$$A_v = 1 + \frac{9 \text{ k}\Omega}{1 \text{ k}\Omega} = 10$$

3.4 Frequency Response Measurement and Gain Calculation

3.4.1 Objective

The purpose of this section is to evaluate the frequency-dependent behavior of the voltage gain of the 741 operational amplifier configured as a non-inverting amplifier. By measuring the amplifier's output over a wide frequency range, we aim to determine its gain-bandwidth product and identify the -3 dB cut-off point, beyond which the amplifier no longer maintains its expected gain performance.

3.4.2 Measured Data During Lab

Table 1: Frequency Response for Gain 20 dB

Frequency (Hz)	U_{in} (mV)	U_{out} (mV)	Gain (V/V)	Gain (dB)
1,000	77.6	777	10.01	20.01
2,000	77.8	777	9.99	19.99
5,000	78.1	777	9.95	19.96
10,000	78.2	773	9.88	19.90
20,000	77.4	754	9.74	19.77
50,000	69.0	607	8.80	18.89
100,000	49.5	328	6.63	16.43
200,000	68.8	270	3.92	11.88
500,000	70.2	132	1.88	5.48
1,000,000	102.0	16.1	0.16	-16.04
2,000,000	32.0	26.0	0.81	-1.80

3.4.3 Logarithmic Graph for Frequency Response

The following graph displays the gain in dB as a function of the input signal frequency on a logarithmic scale. The frequency at which the gain drops by 3 dB from its low-frequency value (20 dB) is defined as the cut-off frequency.

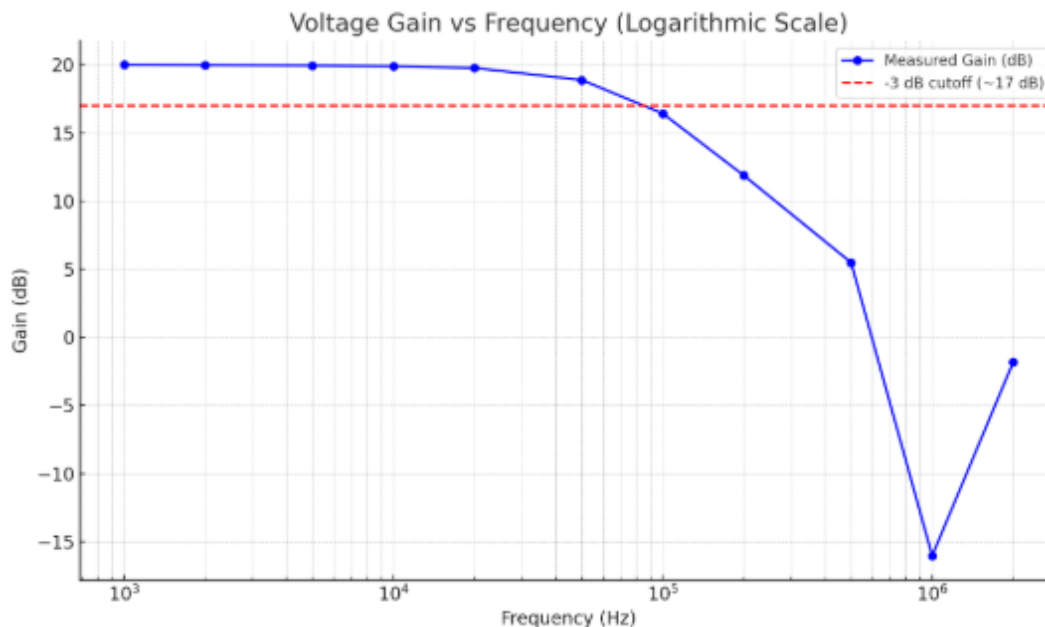


Figure 8: Voltage Gain vs Frequency

3.5 Gain-Bandwidth Product (GBW) Calculation

The Gain-Bandwidth Product (GBW) is a key performance metric of an amplifier, defined as:

$$\text{GBW} = \text{Gain}_{\text{linear}} \times f_{\text{cutoff}}$$

Where:

- $\text{Gain}_{\text{linear}}$ is the voltage gain in linear terms (not dB),
- f_{cutoff} is the -3 dB bandwidth, i.e., the frequency at which the gain drops by 3 dB from its maximum.

In our measurement, the peak gain is approximately 20 dB, which corresponds to a linear gain of:

$$\text{Gain}_{\text{linear}} = 10^{\frac{20}{20}} = 10$$

The -3 dB point corresponds to a gain of:

$$10 \times \frac{1}{\sqrt{2}} \approx 7.07$$

From the measured data, this occurs at approximately:

$$f_{\text{cutoff}} \approx 100 \text{ kHz}$$

Thus, the **Gain-Bandwidth Product** is:

$$\text{GBW} = 10 \times 100 \text{ kHz} = \mathbf{1 \text{ MHz}}$$

This value characterizes the frequency response capability of the amplifier at this specific gain level.

3.5.1 Gain 6 dB Configuration

Table 2: Frequency Response for Gain 6 dB

Frequency (Hz)	Gain (V/V)	Gain (dB)
1,000	1.95	5.82
2,000	1.96	5.84
5,000	1.98	5.89
10,000	2.00	5.91
20,000	2.01	5.93
50,000	2.01	5.95
100,000	1.95	5.84
200,000	1.91	5.76
500,000	1.28	1.08
1,000,000	0.61	-4.30
2,000,000	0.18	-15.10

3.5.2 Gain 40 dB Configuration

Table 3: Frequency Response for Gain 40 dB

Frequency (Hz)	Gain (V/V)	Gain (dB)
1,000	88.88	38.96
2,000	85.80	38.60
5,000	65.74	36.32
10,000	51.39	34.10
20,000	28.94	29.23
50,000	12.33	21.87
100,000	6.52	16.31
200,000	3.07	9.72
500,000	1.28	2.10
1,000,000	0.63	-4.06
2,000,000	0.33	-9.63